Protein synthesis

Translated by Emma Louise Clarkson Abad and Ruben Pio from:

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Learning objectives:

- To understand how genetic information stored in DNA is transferred to an amino acid sequence to form a functional protein.
- To know the molecular aspects of protein synthesis and the elements involved in the process.

INTRODUCTION

The mechanisms by which DNA stores and conserves genetic information, and how RNA transmits that information from the nucleus to the cytoplasm, have been explained in previous chapters. Once in the cytoplasm, the RNA is translated into the amino acid sequence of proteins, which are responsible for cellular functions. The mechanisms of synthesis (**protein translation)** and the post-translational modifications undergone by proteins in eukaryotic organisms are somewhat different from those in prokaryotes.

Proteins are organized sequences of amino acids that form a polypeptide chain and adopt three-dimensional structures. All proteins, both eukaryotic and prokaryotic, are formed from the same set of 20 amino acids (fig. 9-1). Cells can produce proteins with different properties and functions by joining the 20 amino acids in a multitude of different combinations and sequences. In this way, a variety of polypeptides can be produced, such as enzymes, hormones, antibodies, structural proteins of the cytoskeleton, etc. All amino acids have a characteristic structure: they contain a carboxyl and an amino group both bound to the same carbon atom (fig. 9-2). This carbon atom is called the α-carbon and establishes bonds with a hydrogen and a side chain (or R group), which is what differentiates each amino acid from the other, giving them their specific physicochemical properties. To form a protein, amino acids join together to form a substituted amide bond, called a **peptide bond** (see fig. 9-2). This bond allows rotation of the atoms involved, which provides polypeptide chains with three-dimensional flexibility.

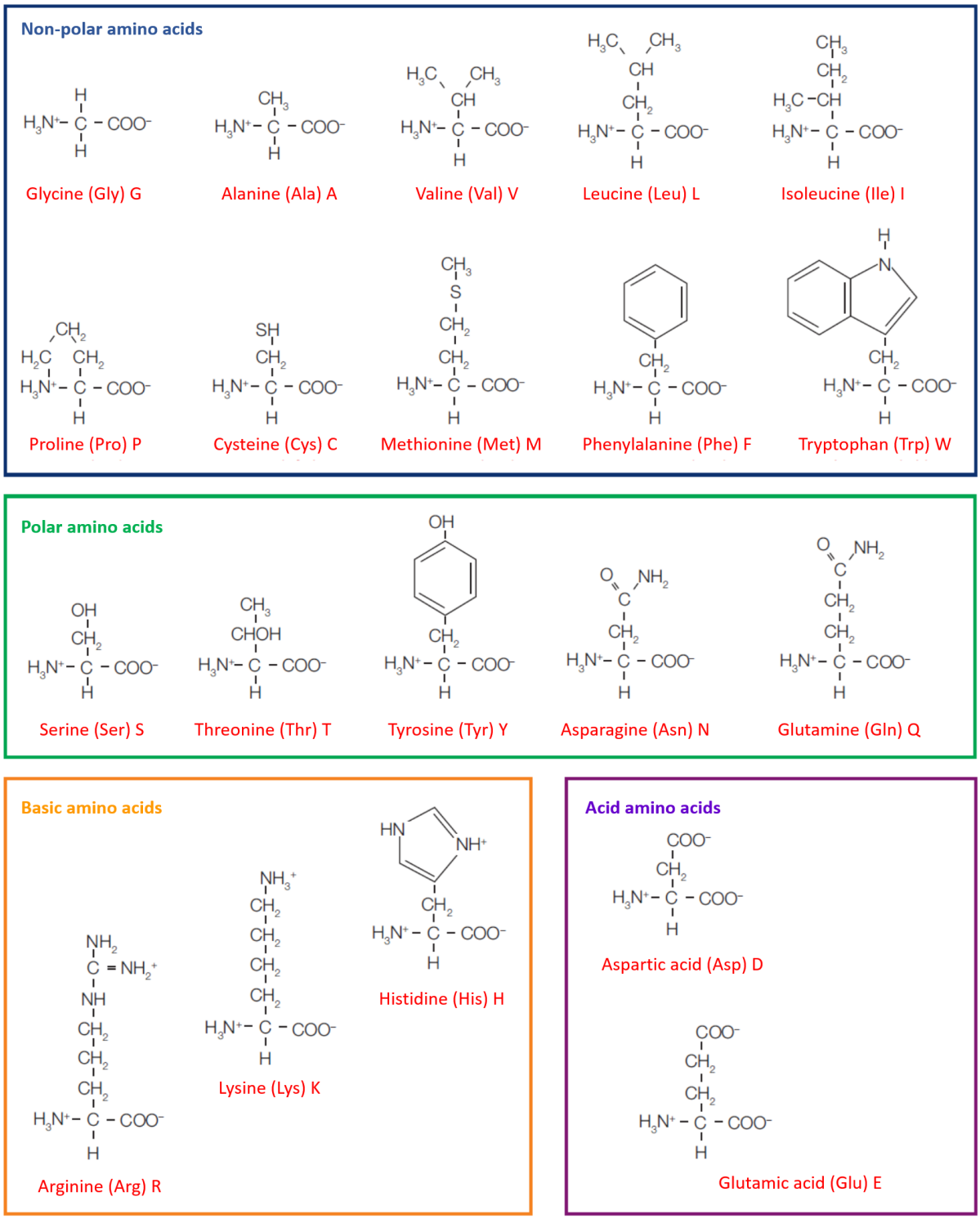


Figure 9-1 Structure of the 20 amino acids. The amino acids inside the blue box correspond to non-polar amino acids, those in the green box are polar, those in the orange box are basic and those in the purple box are acidic. The names of the amino acids are displayed in the three internationally accepted codes: complete name, three-letter code and one-letter code. *(Illustrated by E. Maldonado.)*

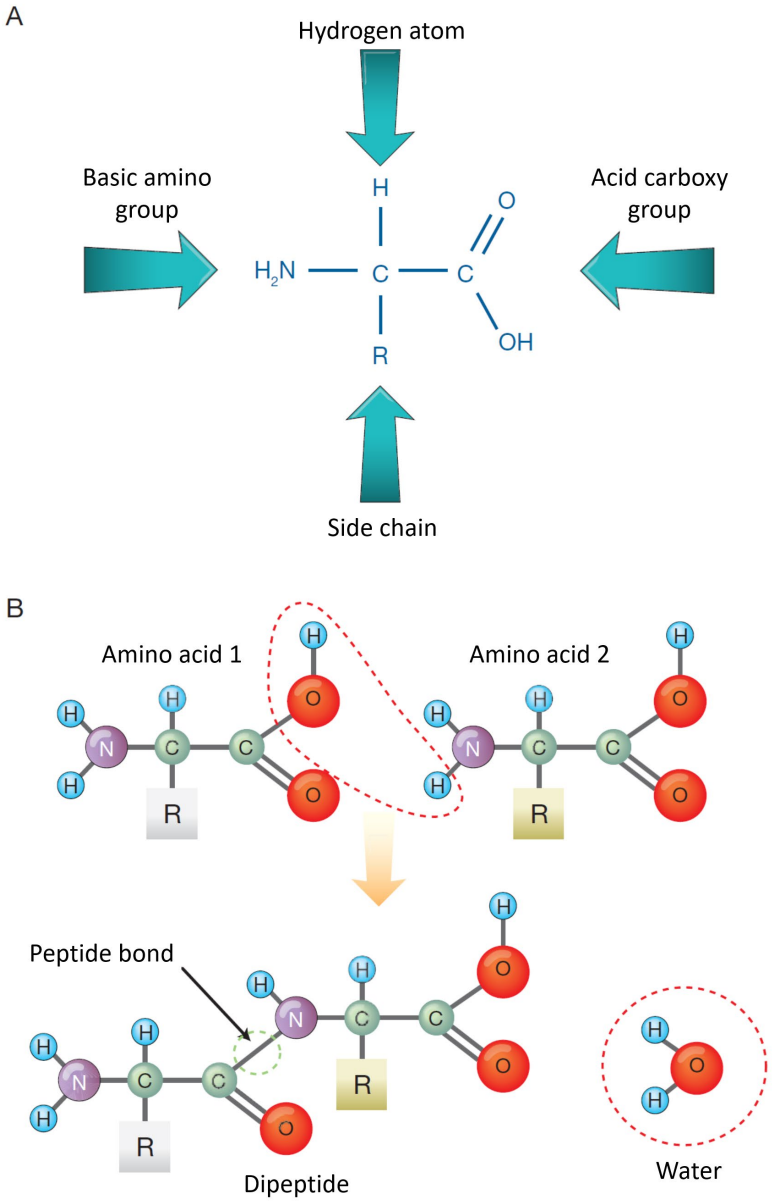


Figure 9-2 **A.** Generic structure of an amino acid. **B.** Formation of a peptide bond between two amino acids. *(Illustrated by E. Maldonado.)*

THE MACHINERY FOR PROTEIN SYNTHESIS

To construct the chain of amino acids characteristic of each protein, three main elements are required: a messenger RNA (mRNA), ribosomes and different transfer RNAs (tRNA), which transport each amino acid to be incorporated into the protein during the synthesis. Ribosomes bind to the mRNA to "read" its genetic information and incorporate tRNAs sequentially, each carrying a specific amino acid, from which the protein is constructed. Typically, several ribosomes simultaneously "read" the genetic information contained in the same mRNA, which results in a highly efficient translation process (fig.9-3). Under a transmission electron microscope, we can observe this set of ribosomes in the process of

translating the same mRNA, forming structures known as **polyribosomes** or polysomes. The number of ribosomes joined to each mRNA depends on the length of the mRNA. On an mRNA of 300 nm, about 10 ribosomes may be present. During translation, ribosomes move along the mRNA strand in a 5' to 3' direction. For every three nucleotides (a codon) that the ribosome moves past, an amino acid is incorporated into the growing polypeptide chain. Once the synthesis process has finished, the ribosomes separate from the mRNA and the newly synthesized protein is released.

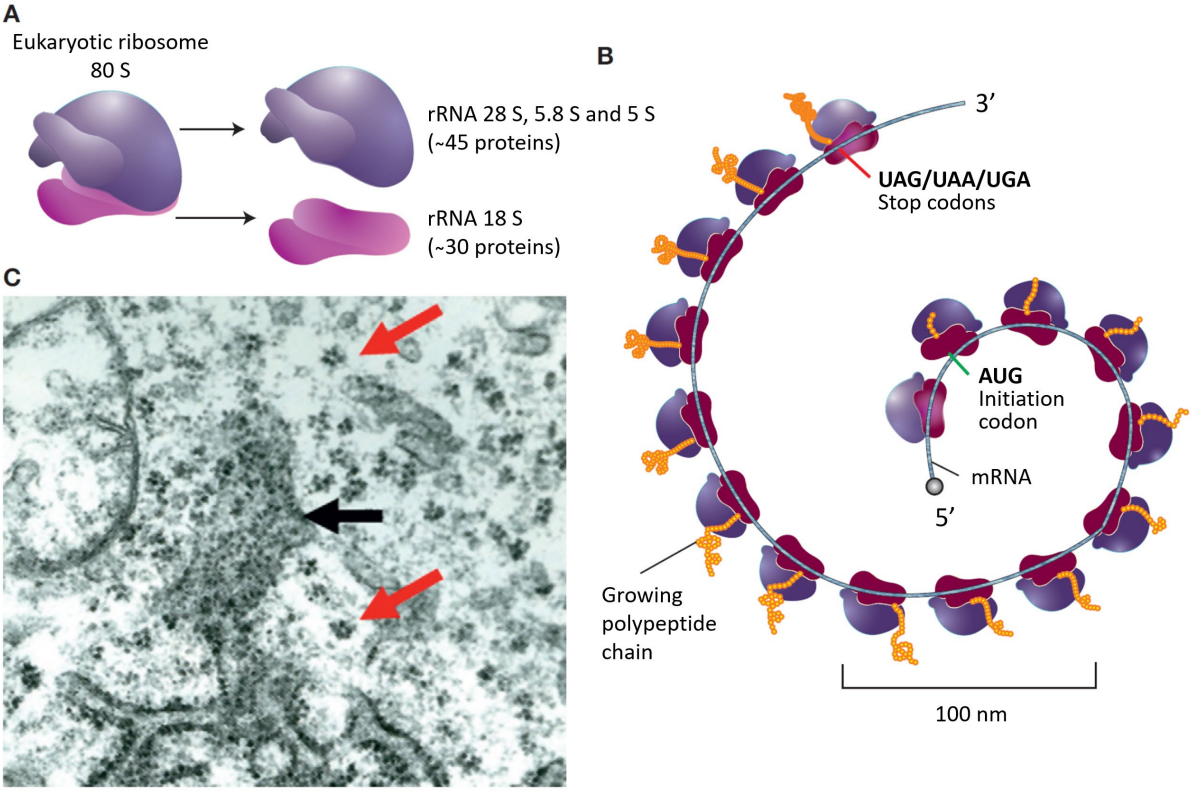


Figure 9-3 Structure of ribosomes and polyribosomes. **A.** Diagram of the structure and subunits of an 80S ribosome. **B.** Representation of a polyribosome (or polysome) where the different ribosomes are shown in the process of translation of the same mRNA strand. **C.** Electron microscope image of polysomes. The black arrow indicates polysomes attached to the surface of the RER; the red arrows indicate polysomes free in the cytoplasm. *(Illustrated by E. Maldonado.)*

Messenger RNA

The mRNA molecule contains, as already indicated, the genetic information about the sequence by which the amino acids are going to be incorporated into the protein. In other words, it is the template used to synthesize the protein. The mRNA consists of a single-stranded sequence with a methylated guanine nucleotide as a cap on its 5' end and a chain of adenines, known as a poly(A) tail (see Chapter 8), on its 3' end. Not all of the mRNA sequence is used as the template for protein synthesis; instead, the coding sequence is limited to the portion between the start and the stop codons.

Ribosomes

Ribosomes are the cellular structures where translation takes place. Ribosomes are ribonucleoproteins formed by various ribosomal RNAs (rRNA) associated with a set of different proteins (see fig. 9-3). Eukaryotic ribosomes have a sedimentation rate of 80S, and are formed by two subunits. The small subunit has a sedimentation rate of 40S and contains a single rRNA of 18S and about 30 proteins. The large subunit has a sedimentation rate of 60 S and contains 3 rRNAs (28S, 5.8S and 5S) and around 45 proteins. Prokaryotic ribosomes have a sedimentation rate of 70S. The large subunit, of 50S, contains one rRNA molecule of 23S and another of 5S, as well as 34 proteins. The small subunit contains one 16S rRNA and 21 proteins. As mentioned in chapter 8, the synthesis of the different rRNAs and the assembly of the ribonucleoproteins that form the ribosome take place in the cell nucleolus. Each one of the subunits formed in the nucleolus (large and small) is exported from the nucleus to the cytoplasm separately. The subunits will assemble on the mRNA when translation is initiated.

Ribosomes can be found free in the cytoplasm or attached to the cytoplasmic side of the rough endoplasmic reticulum (RER), because many proteins are synthesized while being incorporated into the interior of this organelle. As covered in the section “Transport and localization of proteins”, the fact that a protein is synthesized in the cytoplasm or in the RER will determine its final destination in the cell. There are also ribosomes in the mitochondria, since this organelle can perform its own protein synthesis (see chapter 11). However, in this case, the ribosomes are different from the cytoplasmic ones. They have a sedimentation rate of 55-60S, with a large subunit of 35S (which contains an rRNA of 16S, similar to prokaryotic ribosomes, and another of 4S) and a small subunit of 25S (with a 12S rRNA). Mitochondrial ribosomes are similar to those of prokaryotic organisms not only in their structure, but also in their sensitivity towards some antibiotics or other drugs that inhibit translation (see chapter 11).

Transfer RNA

tRNAs are the molecules responsible for providing the necessary amino acids to construct proteins (see chapter 8). tRNAs are relatively small RNAs (4S), formed by a single RNA strand folded in a three-dimensional manner, which is usually described as a cloverleaf-like structure (fig. 9-4). tRNAs have between 73 and 93 nucleotides and are organized in four arms, which are crucial to their function. On one side of the molecule, the 3’ end of the tRNA, which always ends in the trinucleotide sequence **CCA**, binds to the amino acid. The amino acid remains linked by an ester bond between its carboxyl group and the 2’ or 3’ hydroxyl group of the terminal adenosine residue (A) on the **amino acid arm**. On the other side of the molecule, the **anticodon arm** (found on the opposite end to the arm that binds the amino acid) has the specific complementary trinucleotide sequence to a codon on the mRNA strand, which determines the amino acid associated with each codon. Each anticodon (and subsequently each tRNA) is specific to one amino acid, although as will be explained later, some amino acids are recognized by and can bind to two or more tRNAs. Another of the tRNA arms will be responsible for attaching the tRNA to a ribosome. This arm is known as the **T loop,** because it contains the nucleoside T (ribothymidine), which forms part of the T-ψ-C sequence (ψ is an uncommon nucleoside, known as pseudouridine). Finally, there is an arm in the tRNA that recognizes the enzyme aminoacyl-tRNA synthetase (see further). This arm is known as the **D loop**, because it contains two D nucleosides (dihydrouridine).

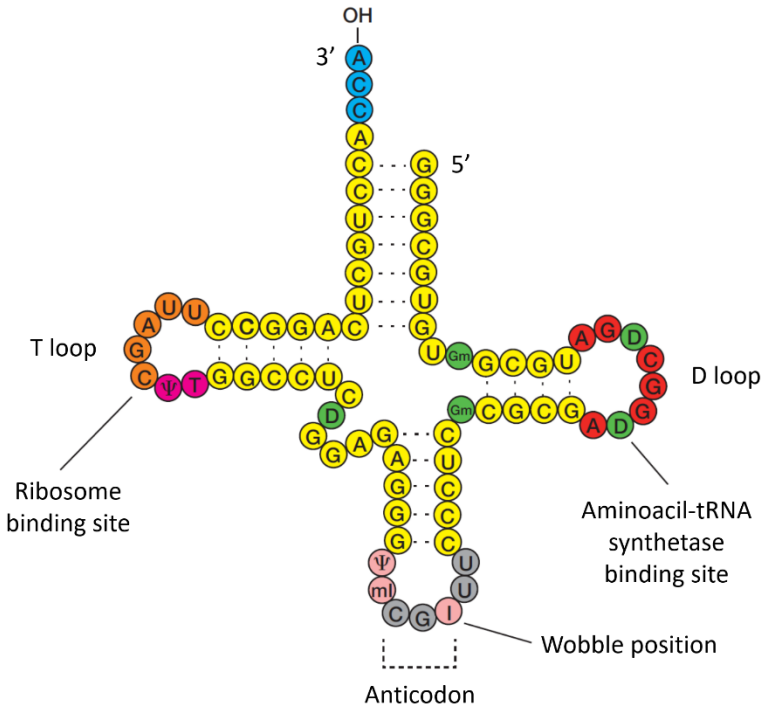


Figure 9-4 Diagram of the structure and different parts of a tRNA. D is dihydrouridine, T is ribothymidine, Gm is methylguanosine, I is inosine, ml is methylinosine and ψ is pseudouridine. *(Illustrated by E. Maldonado.)*

THE GENETIC CODE

Once the elements necessary for protein synthesis have been described, an important question arises: How is the genetic information encoded in a language of four nucleotide "letters", A, C, G and T (U in the RNA) in nucleic acids translated to a language of 20 amino acid "letters" in proteins? The answer lies in the combination of the four nucleotides. The four nucleotide "letters" associated in pairs would only produce 16 combinations (4^2), which is an insufficient number to cover all the amino acids (20); therefore, the solution has to be in the arrangement in groups of three, resulting in 64 possible combinations (4^3). Groups of three nucleotides in mRNA are known as codons. The sequence of each codon is complementary to that of an anticodon present on a particular tRNA, which in turn is associated with a specific amino acid. The mRNA is "read" by the translation mechanism as a sequence of codons. Each codon will determine the incorporation of an amino acid into the protein, such that the sequence of codons on the mRNA will be translated to a sequence of amino acids in the protein. This code of information is known as the **genetic code** (fig. 9-5).

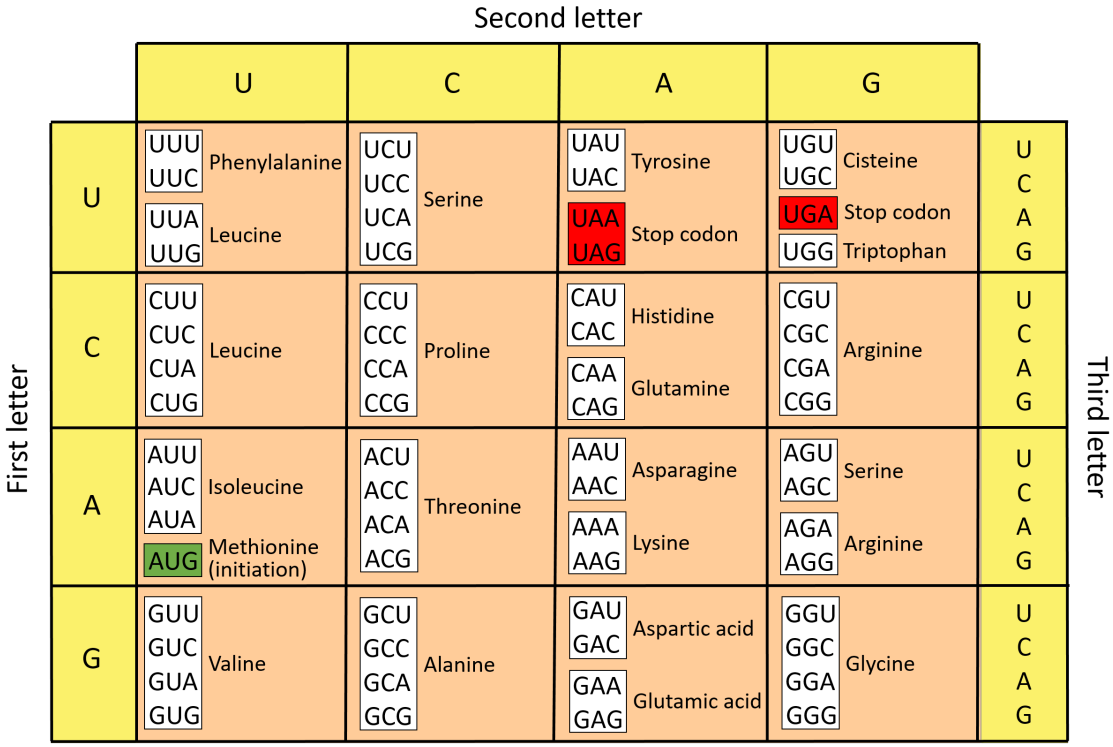


Figure 9-5 The genetic code. The triplets of the mRNA indicate which amino acid is incorporated into the protein in translation. Some codons mark the end of translation.

As can be observed, there are more available codons than possible amino acids. This provides the genetic code with a fundamental characteristic. The genetic code is **degenerate**, which means that a given amino acid is encoded by more than one codon. It is important to note that degenerate does not mean imperfect; the genetic code is non-ambiguous, because no one codon will code for more than one amino acid. When an amino acid has more than one codon to code for it, the difference between these codons usually lies in the third base. The first two letters of each codon are, therefore, primary determinants of the specificity. The rules governing the specificity of the binding of a codon and an anticodon in a tRNA were established by Francis Crick, and were set out as follows:

- The first two bases of a codon form strong bonds with their corresponding pairs, i.e. the two last bases on the anticodon.
- When the first base of the anticodon is C or A, the binding of the third base of the codon is specific and only one codon is recognized by the tRNA. When, however, the first base of the anticodon is U or G, the binding to the third base is less specific and two codons can be recognized by the tRNA. Finally, when inosine (a modified nucleotide present in tRNA) is the first base of the anticodon, the binding with the third nucleotide of the codon tends to be very weak and the tRNA can recognize up to three codons.
- When an amino acid is encoded by various codons, the codons which differ in either of the two first bases require different tRNAs.

Bearing in mind the previous rules, a minimum of 32 tRNAs are required to translate the 61 codons of the genetic code. The three remaining codons are not recognized by any tRNA and are known as **termination codons,** because they signal the end of the synthesis of the protein chain.

THE SYNTHESIS OF PROTEINS IN RIBOSOMES

Once the essential elements required for the synthesis of proteins and the necessary mechanisms for "reading" the genetic information contained in the mRNA have been explained, we can proceed to explain the translation process which takes place in the ribosome. Notably, although the ribosomes, mRNA and tRNA molecules are three key elements for this function, the translation also involves the participation of many other protein factors that are essential for the process. These factors are known as initiation, elongation and termination factors. For protein synthesis to occur, each tRNA needs to pair up first with its corresponding amino acid. The steps of this process are as follows (fig. 9-6, *A*):

- Activation of an amino acid by its binding to AMP, forming an intermediate aminoacyl-AMP. This reaction requires ATP.
- Binding of the aminoacyl-AMP to the tRNA on the terminal sequence CCA of the amino acid arm, releasing the AMP and forming an aminoacyl-tRNA.

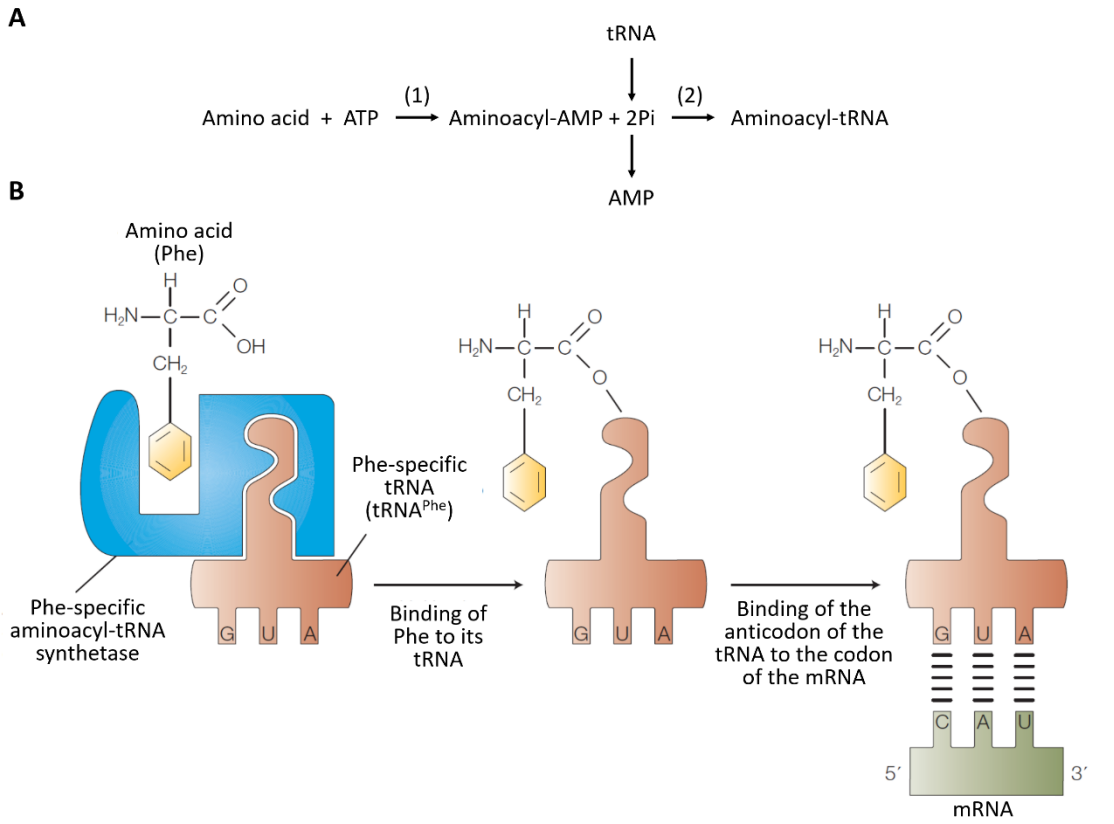


Figure 9-6 Each tRNA carries a specific amino acid. **A.** The binding process starts with the formation of an aminoacyl-AMP (1) and the subsequent formation of an aminoacyl-tRNA (2). These reactions involve the actions of aminoacyl-tRNA synthetases, with a corresponding energy consumption. B. The aminoacyl-tRNA synthetase specific for phenylalanine is shown as an example of how a phenylalanine molecule binds to its corresponding tRNA. *(Illustrated by E. Maldonado.)*

Both reactions are catalyzed by an aminoacyl-tRNA synthetase specific for each amino acid. The specific recognition is possible due to the active center of these enzymes, where only a specific amino acid can fit. Therefore, there are (at least) 20 different aminoacyl-tRNA synthetases to recognize each one of the 20 amino acids (see example shown in fig. 9-6, *B*). In parallel with the binding of amino acids to their corresponding tRNAs, the small ribosomal subunit attaches to the mRNA to be translated. Following the association of the large subunit, the different tRNAs can enter a specific site on the ribosome (see the process described in greater detail below). Meanwhile, the large subunit of the ribosome is in charge of catalyzing the formation of peptide bonds in the growing polypeptide.

The two ribosomal subunits have internal domains which accommodate the tRNAs throughout the different phases of protein synthesis (fig. 9-7, *A*). The **A** site refers to the binding site of the aminoacyl-tRNA that initially enters the ribosome. The **P** site refers to the space occupied by the aminoacyl-tRNA once it has been bound to the growing polypeptide chain (peptidyl-tRNA). The **E** site corresponds to the exit site of the tRNA that has transferred the growing polypeptide chain and has to leave the ribosome. The translation process can be divided into three defined steps:

Initiation

During this phase, the small ribosomal subunit will bind to the mRNA (figs. 9-7, *B* and 9-8, step 1). Translation in eukaryotic cells always starts at an **AUG** codon, which codes for methionine (Met). The **initiating tRNA** (which is bound to the amino acid methionine) binds to the small ribosomal subunit on the P site, aided by the initiation factor eIF2 (*eukaryotic initiation factor 2)* bound to GTP. Meanwhile, the small subunit is associated with eIF1, eIF1A, eIF3 and eIF5. The mRNA also needs to bind to several other factors: eIF4E, eIF4G (both associated with the 5' region of the mRNA), eIF4A, eIF4B and PABP (*poly(A) binding protein*). The binding of this protein allows the 3' region of the mRNA (which contains the poly(A) tail) to fold back and come close to the 5' region.

After binding, the **pre-initiation complex** formed by the small ribosomal subunit and the aforementioned factors moves along the mRNA (through the 5' UTR region, *untranslated region*) until the AUG initiation codon is located over the P site on the ribosome (see fig. 9-8, steps 2 and 3). This process requires a large expenditure of energy in the form of ATP. When the initiation codon is recognized by the anticodon, eIF5 induces the hydrolysis of the GTP present on eIF2. Then, the initiation factors (including eIF2, which is now bound to GDP) are released and a new factor called eIF5B-GTP (not shown in the diagram) allows the association of the large subunit coupled to the hydrolysis of GTP (fig. 9-8, step 4). In this way, the machinery for protein synthesis is ready for the next step (see fig. 9-8).

Elongation

The entry of the second tRNA (for example Phe-tRNA^{Phe}) into the A site of the ribosome (fig. 9-9, step 5) requires the assistance of two *eukaryotic elongation factors* (eEF1A, bound to GTP, and eEF1B). The correct positioning of the aminoacyl-tRNA on the A site produces a conformational change that induces the hydrolysis of the GTP associated with eEF1A, which causes the release of this factor (now bound to GDP). Afterwards, the amino acid bound to the tRNA moves towards the P site, where the first amino acid incorporated into the chain

(Met) awaits. The orientation of both of these amino acids on the ribosome allows the formation of a peptide bond catalyzed by the **peptidyl-transferase activity** found in an rRNA of the large ribosomal subunit, which works like a ribozyme (see chapter 8). The polypeptide chain formed by the first two amino acids remains anchored to the tRNA placed on the A site (see fig. 9-9, step 6). After the formation of the first peptide bond, the ribosome moves in a 5' to 3' direction on the mRNA, which causes the tRNA bearing the amino acids (Met and Phe) to be sited now on the P site, and the tRNA that initially carried Met on the exit site (E). In this way, the next available codon of the mRNA strand is located on the A site. This movement is referred to as translocation and requires energy, which is provided by the hydrolysis of a GTP molecule bound to the eEF2 factor. This factor binds to the ribosome once the catalysis of the peptide bond has taken place. Apart from providing the necessary energy for translocation, this factor also contributes to translocation through conformational changes coordinated with the ribosome. When translocation has finished, eEF2 and the tRNA on the E site dissociate from the ribosome, which is now prepared for a new elongation cycle.

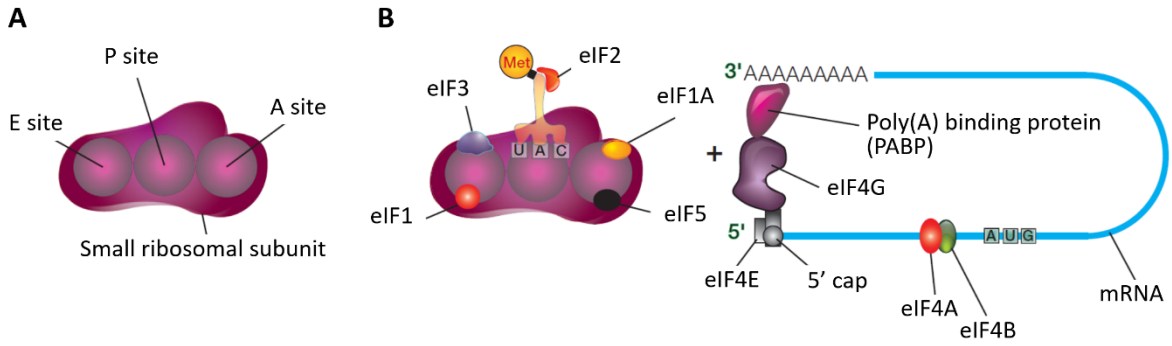


Figure 9-7 **A.** Sites of the small ribosomal subunit. E (exit), P (peptidyl) and A (aminoacyl). **B.** Initiation factors that bind to the initiating tRNA, the small ribosomal subunit and the mRNA to start the process of translation. *(Illustrated by E. Maldonado.)*

Termination

As elongation goes on, new mRNA codons become available on the A site and new amino acids are added to the polypeptide chain being formed (fig 9-9, steps 7 and 8) until a **stop codon** (UAA, UAG or UGA) reaches the A site (fig.9-10, step 9). The stop sequences in eukaryotes are recognized by a specific domain of the **termination factor** eRF3 (*eukaryotic release factor 3*), which forms a complex with another termination factor, eRF1. Once this protein complex has bound to the ribosome, a conformational change takes place in the complex, which allows eRF1 to induce the hydrolysis of the bond between the newly synthesized protein and the tRNA, releasing the protein (see fig. 9-10, steps 10 and 11). Subsequently, the recently formed protein and the termination factor detach from the ribosome, ending the translation process. Once the synthesis has been concluded, the tRNA and the mRNA remain bound to the ribosome. The aid of a protein complex formed by various factors will be required for their excision, although the mechanism by which this process occurs is mostly unknown.

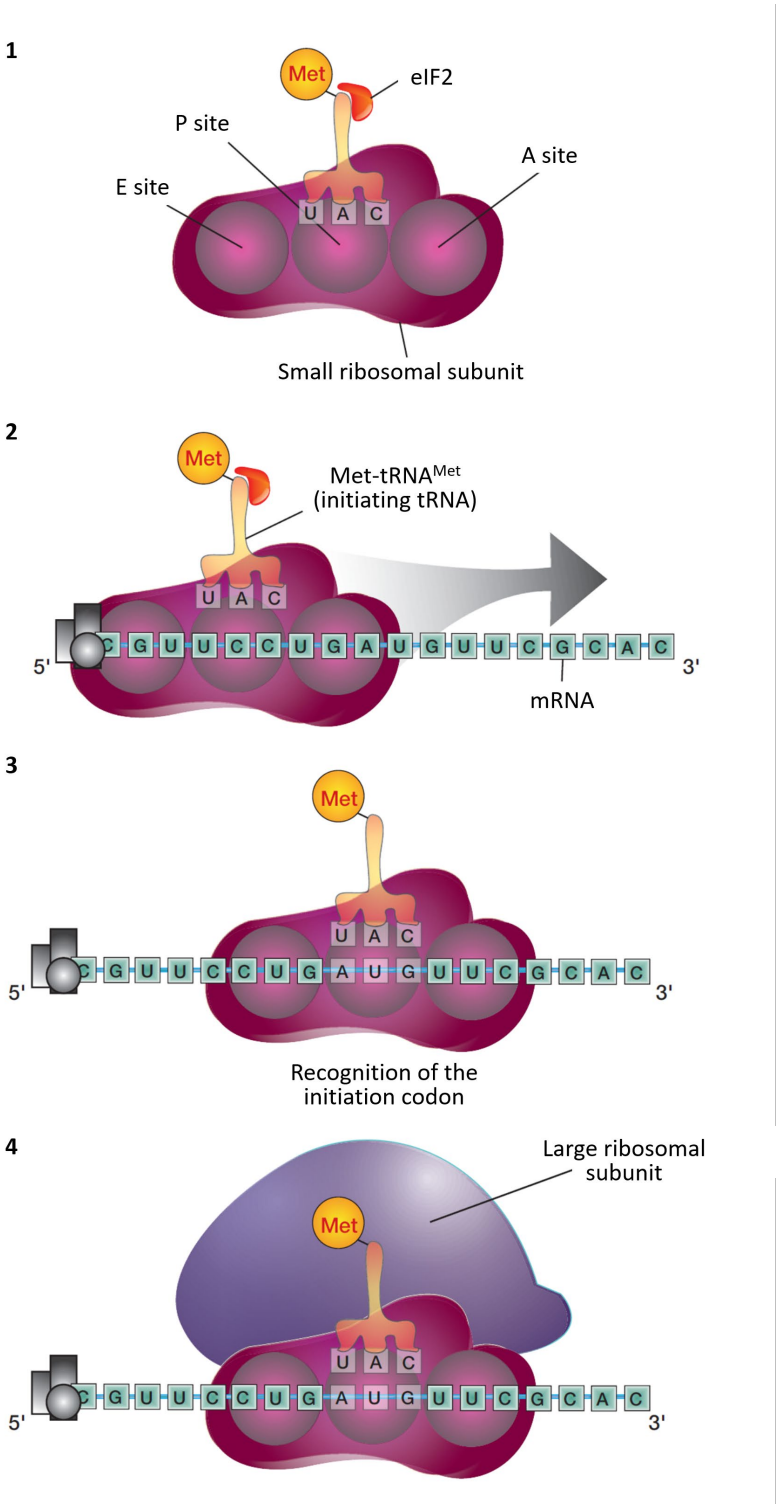


Figure 9-8 Initiation of translation. The small ribosomal subunit binds to the initiating tRNA (carrying the amino acid methionine) (1) and moves along the mRNA (2) until it recognizes the start codon AUG, complementary to the anticodon on the tRNA (3). This binding involves the coupling of the large ribosomal subunit (4). *(Illustrated by E. Maldonado.)*

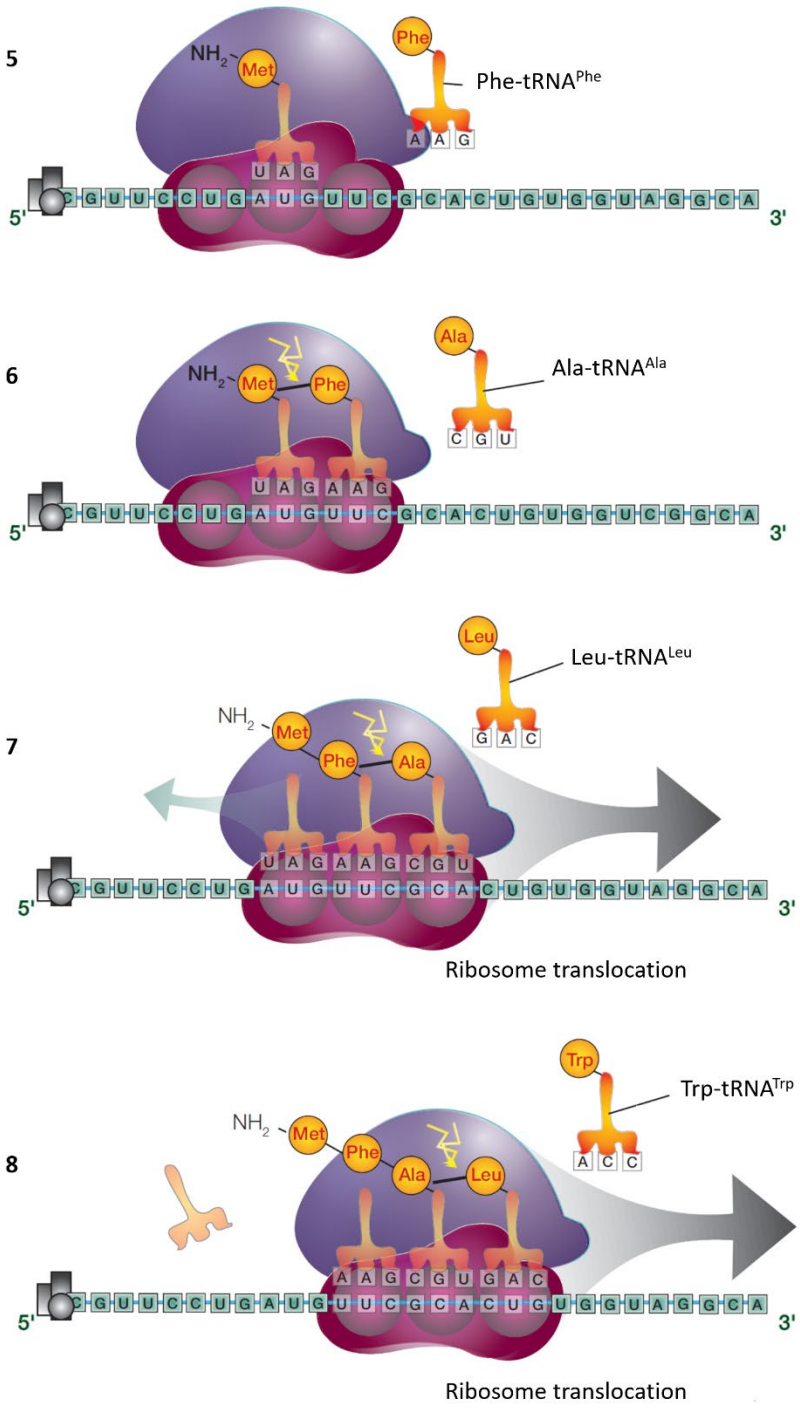


Figure 9-9 Elongation in translation. A new tRNA is introduced, carrying a new amino acid (in this case phenylalanine, Phe), on the A site and a peptide bond between methionine (Met) and Phe on the A site (5, 6). Afterwards, the ribosome moves in a 5' to 3' direction and the tRNA carrying the dipeptide is positioned on the P site, while the tRNA that initially carried Met is positioned at the E site. A new amino acid is introduced (alanine, Ala) on the A site. The process repeats itself with the incorporation of new amino acids, completing the synthesis of the polypeptide chain (7, 8).

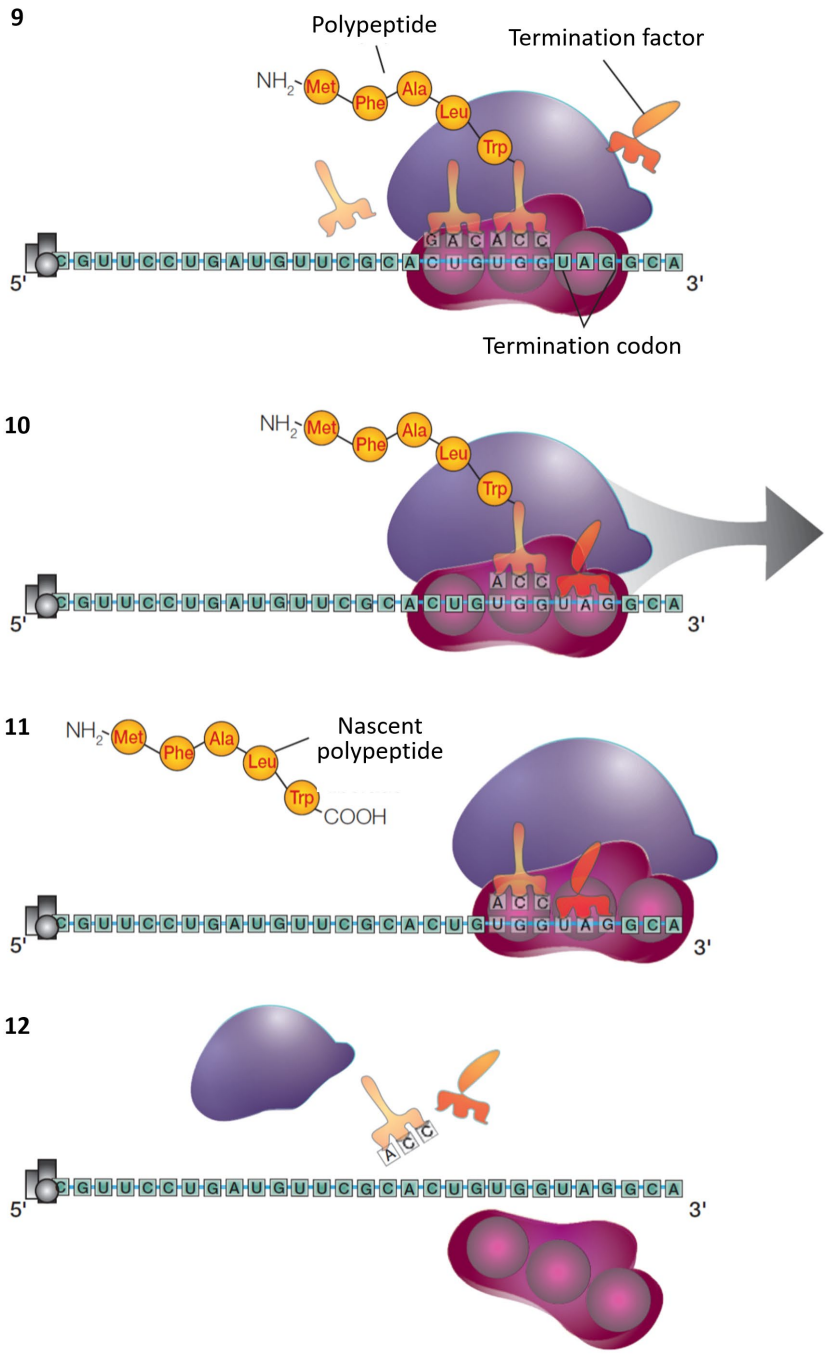


Figure 9-10 Termination of the translation process. The presence of a stop codon on the A site determines the binding of a termination factor (9, 10), which contributes to the release of the polypeptide (11) and the disassembly of the ribosome (12). *(Illustrated by E. Maldonado.)*

The translation of proteins can be interrupted by the use of certain chemical compounds, such as **puromycin**, whose structure is similar to that of a tRNA molecule bound to an amino acid. **Cycloheximide** is another compound that blocks translation, because it inhibits the translocation of 80S ribosomes. This chemical compound is frequently used in biochemical experiments to study cellular processes in which the absence of translation is required.

Protein synthesis is a vital process for all organisms, which explains why diseases associated with defects in the mechanisms of protein synthesis are extremely rare. Any significant alteration in the components of translation would often result in immediate death of the individual, before even entering the embryonic phase of development. Nevertheless, there are some diseases associated with errors in the structure of proteins that participate in translation. An example would be the **Diamond-Blackfan anemia**, an autosomal dominant congenital disorder that manifests in the neonatal period. Affected individuals suffer from erythroid aplasia (a marked reduction in red blood cell production). Half of the patients suffer from additional symptoms that include urogenital or craniofacial deformities, abnormal extremities and cardiac defects. Although this pathology has been linked to various molecular alterations, in most cases, it is due to a mutation of the *RPS19* gene, which codes for different isoforms of a protein (88 to 144 amino acids in size), which is integrated into the small ribosomal subunit. The mechanism by which the alteration of this protein interferes with the normal development of red blood cells and is associated with other developmental abnormalities has not yet been determined.